Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an energy-transfer model and communicate its limits.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain why a model result depends on its assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “internal energy” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

internal energy: energy stored in a system’s microscopic motion/interactions

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A heat engine absorbs Q_{hot}=2400 J and rejects Q_{cold}=1680 J each cycle.
Find work and efficiency.

Word help: cycle = a process that returns a system to its initial state

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

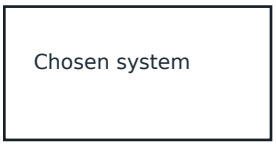
Reference: $Q = mc\,\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. Select the applicable model and supplied constants. State heat-loss, phase-change and cycle assumptions.

Worked reasoning: A system takes 120 J of heat, does 50 J of work and increases internal energy by 70 J. An energy model must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A hot block and cool water are isolated together.
Draw initial heat flow and describe equilibrium without inventing a final temperature.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Hot block

Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: 1 2 3

04 Apply the physics

A beaker contains 0.10 kg of ice at 0°C. Melt it and warm the water to 10°C.
Use L_f = 334,000 J/kg and c_{water} = 4200 J/(kg·°C).
Show both stages in the energy account and find the total energy required.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

An engine absorbs 500 J and produces 150 J work in a cycle.
Find rejected heat and explain why an energy balance alone does not establish that any temperature pair can achieve that efficiency.

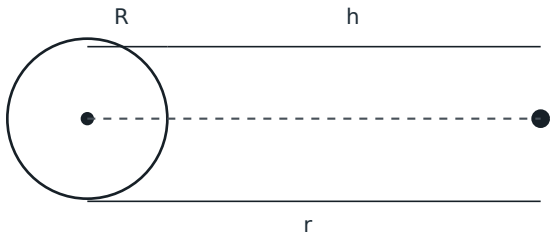
Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simulated source has GM=288 m³/s². At r=4 m, find escape speed and state two assumptions.
Label the distances or energy levels used in your reasoning.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A heat engine between 500 K and 300 K absorbs 1000 J.
Find ideal maximum work and explain why actual work usually falls below it.
Label energy entering, leaving or stored on the system model.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg sample with c=400 J/(kg·K) warms 2 K while doing 60 J of work on surroundings.
Find heat input under $\Delta U = mc\,\Delta T$ and state one condition required for that model.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3